

“You Do It” Solution

Lecture 8: Hypothesis Testing: One-sample Tests

Prepared as a classroom-ready detailed solution set for ECOM5005

Exercise 1: Interpreting a p -value

Question

When asked to explain the meaning of “the p -value was $p = 0.03$ ”, a student says, “This means there is only probability 0.03 that the null hypothesis is true.” Is this an essentially correct explanation? Explain your answer.

Detailed Solution

Step 1: Identify the student’s interpretation.

The student is treating the p -value as

$$P(H_0 \text{ is true} \mid \text{observed data}) = 0.03.$$

That is **not** what a p -value measures.

Step 2: State the correct definition.

A p -value is calculated **assuming that H_0 is true**. It is the probability of obtaining a test statistic at least as extreme as the one actually observed. If E denotes “a result at least as extreme as the observed result”, then

$$p = P(E \mid H_0).$$

Thus, $p = 0.03$ means $P(E \mid H_0) = 0.03$.

Step 3: Explain the error in the student’s reasoning.

The student has reversed the conditioning. The p -value is $P(E \mid H_0)$, whereas the student’s statement is about $P(H_0 \mid E)$. In general,

$$P(E \mid H_0) \neq P(H_0 \mid E).$$

So $p = 0.03$ does **not** mean that H_0 has a 3% chance of being true, nor that H_0 has a 97% chance of being false.

Step 4: Give the correct verbal interpretation.

If the null hypothesis is true, there is a 3% probability of obtaining a test statistic at least as extreme as the one observed.

The sample evidence would therefore be relatively unusual under H_0 .

Step 5: Link the p -value to the test decision.

Compare p with the chosen significance level α : if $p < \alpha$, reject H_0 ; otherwise, do not reject H_0 . Hence, $p = 0.03$ implies rejection at $\alpha = 0.05$, but not at $\alpha = 0.01$.

Exercise 1: Interpretation and Final Answer

Why the Student's Explanation Is Incorrect

The explanation is **not correct**. A p -value is computed by first assuming that the null hypothesis is true and then asking how unusual the observed sample result would be under that assumption. For $p = 0.03$,

Assume H_0 is true $\rightarrow$ only a 3% chance of data this extreme or more extreme.

The probability statement therefore runs from the hypothesis to the possible sample evidence. It does **not** run in the reverse direction from the observed evidence to the probability that the hypothesis is true.

Conditional-Probability Distinction

The key distinction can be written compactly as

$$\underbrace{P(E \mid H_0)}_{p\text{-value}} = 0.03 \quad \text{versus} \quad \underbrace{P(H_0 \mid E)}_{\text{student's claim}} = 0.03.$$

These two conditional probabilities answer different questions. The p -value supplied by a classical hypothesis test gives the first quantity, not the second.

Answer Summary

Final answer: No, the student's explanation is not essentially correct.

A p -value of 0.03 means that, **if the null hypothesis were true**, the probability of observing a test statistic at least as extreme as the one obtained is 0.03, or 3%.

It does **not** mean

$$P(H_0 \text{ is true} \mid \text{data}) = 0.03.$$

If the chosen significance level is $\alpha = 0.05$, then $0.03 < 0.05$ and we reject H_0 . The statistical decision comes from comparing the p -value with α ; the p -value itself is not the probability that H_0 is true.

Final Check

For interpreting a p -value, remember:

- begin with the assumption that H_0 is true;
- the p -value measures how unusual the observed result, or a more extreme result, would be under H_0 ;
- a small p -value provides evidence against H_0 , but is not the probability that H_0 is true;
- compare p with the chosen significance level α : if $p < \alpha$, reject H_0 ; otherwise, do not reject H_0 .